



THE UNIVERSITY OF ZAMBIA

**SCHOOL OF NATURAL AND APPLIED SCIENCES
DEPARTMENT OF COMPUTING AND INFORMATICS**

Final Year Project Proposal

Academic Year: 2026

**Project Title: MOBILE-BASED SYSTEM FOR CONNECTING STUDENTS WITH NEARBY TUTORS
FOR PHYSICAL LESSONS**

Student Name(s)

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Supervisor: MR Martin PHIRI

Department: Computing and Informatics

Date of Submission:

1. Introduction

Access to quality education remains a challenge for many students due to geographical barriers, transportation costs, and limited availability of nearby tutors. While digital learning platforms exist, they primarily focus on online lessons, which may not be effective for all learners, especially those who benefit from face-to-face interaction.

This project proposes a Mobile-Based System for Connecting Students with Nearby Tutors for Physical Lessons within the same geographical area. The system facilitates the scheduling of physical, one-on-one tutoring sessions, reducing the need for long-distance travel and improving accessibility to personalized education.

1.1 Problem Statement

Many students struggle to access academic support due to long travel distances and the high cost of transportation. Existing platforms emphasize virtual learning, which may not suit all educational needs. There is currently no efficient system that connects students with tutors within their immediate locality for physical learning sessions.

1.2 Project Aim

To design and implement a mobile-based system that matches students with nearby tutors and enables the scheduling of in-person tutoring sessions.

1.3 Objectives

1. Develop a system that matches students with tutors based on geographical location.
2. Enable scheduling and booking of physical one-on-one tutoring sessions.
3. Provide tutor profiles including subjects, availability, and ratings.

1.4 Scope of the Project

1.4.1 In Scope

- Location-based tutor-student matching
- Tutor profiles and subject listings
- Booking and scheduling system
- Notification system for appointments
- User registration and authentication
- Rating and feedback system

1.4.2 Out of Scope

- Online/virtual classes (video calls)
- Pre-recorded tutorials or learning materials
- Multi-platform deployment beyond Android (initial version)

2. Literature Review

Location-based and matching systems have been widely used in different domains to connect users efficiently.

- Uber demonstrates how users are matched with nearby service providers using geographic data.
- Google Maps provides location-based services that enable distance calculation and route optimization.
- Preply connects students with tutors, though primarily for online sessions.
- UNESCO (2023) emphasizes the importance of improving access to education through innovative delivery methods.

These systems highlight the effectiveness of location-based matching, which can be adapted to support physical tutor-student interactions.

3. Proposed Methodology

The project will follow an iterative and incremental development approach, allowing continuous improvement throughout development.

3.1 Procedural Workflow

- 1. Requirement Analysis**
Identify user needs, system features, and geographical constraints.
- 2. System Design**
Design system architecture, UI/UX, and matching logic.
- 3. Database Design**
Create database structure for users, tutors, bookings, and ratings.
- 4. Development / Implementation**
 - Mobile application development
 - Backend integration

5. **Testing**

Perform unit testing, integration testing, and user acceptance testing.

6. **Deployment**

Deploy the application and backend services.

7. **Maintenance**

Continuous updates and system improvements.

3.2 Tools and Technologies

- Programming Languages: Dart (Flutter) / Java
- Frameworks: Flutter / React Native
- Database: Firebase / MySQL
- Development Tools: Android Studio, Visual Studio Code
- Technologies: GPS / Location Services

3.3 Ethical and Legal Considerations

- Ensure user privacy and secure handling of personal data
- Obtain user consent for location tracking
- Implement secure authentication mechanisms
- Prevent misuse of the platform

4. Expected Outcomes

- A functional mobile system for tutor-student matching
- Improved accessibility to nearby tutors
- Reduced travel costs and time
- Enhanced learning through personalized sessions
- Complete system documentation

5. Project Timeline

Phase	Duration
Proposal & Requirements	Week 1–2
Design & Planning	Week 3–5
Development	Week 6–10
Midway Evaluation	Week 11
Testing	Week 12–13
Report Writing	Week 14–15
Final Submission	Week 16

Student Names and Responsibilities (Balanced Contribution)

1. Patience MAKAYI

Project Coordination, System Design, Progress Tracking Module

2. Melody TEMBO

Mobile Application Development, Deployment

3. Samuel PHIRI

Database Design, Backend Development

4. Christopher KUMWENDA

Testing, Quality Assurance, Documentation

5. Solomon MUSYANI

UI/UX Design, Content Management

6. Risk Analysis

Risk	Mitigation Strategy
Inaccurate location data	Use reliable GPS services
Low user adoption	Design simple and user-friendly UI
Data privacy concerns	Implement encryption and authentication
Development delays	Weekly progress tracking
Matching inefficiency	Optimize location-based algorithms

7. References

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- [3] UNESCO, *Policy Guidelines for Mobile Learning*. Paris, France: UNESCO, 2013.
- [4] Moodle Mobile, "Moodle App Documentation," 2022. [Online]. Available: <https://docs.moodle.org>
- [5] Khan Academy, "About Khan Academy," 2023. [Online]. Available: <https://www.khanacademy.org>
- [6] H. K. Kim and J. M. Park, "A location-based mobile tutoring system using GPS technology," *International Journal of Computer Applications*, vol. 176, no. 25, pp. 15–2

Supervisor Approval

Name:

Signature:

Date: